College of Science and Technology

Royal University of Bhutan

Computing Technologies Department

Bachelor of Engineering in Information Technology Autumn Semester 2026

CTE412 – Integrated System

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Std. no: 02230129

Task 1

1. Create a kind cluster named cte412-<sid> with one control-plane node and two worker nodes. Host port 30090, bound to 127.0.0.1 only, must reach node port 30090 on the cluster.

```yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
name: cte412-02230129
nodes:
  - role: control-plane
    extraPortMappings:
      - containerPort: 30090 # NodePort inside the kind node
        hostPort: 30090 # published on the Windows host
        listenAddress: "127.0.0.1" # loopback only, per brief
        protocol: TCP
  - role: worker # notice-board pods
  - role: worker # pod spread + multi-pod /version responses

```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab
14> kind create cluster --name cte412-02230129 --config cluster/kind-cluster.yaml
Creating cluster "cte412-02230129" ...
 • Ensuring node image (kindest/node:v1.37.0)  ...
 ✓ Ensuring node image (kindest/node:v1.37.0)
 • Preparing nodes    ...
 ✓ Preparing nodes
 • Writing configuration   ...
 ✓ Writing configuration
 • Starting control-plane   ...
 ✓ Starting control-plane
 • Installing CNI   ...
 ✓ Installing CNI
 • Installing StorageClass   ...
 ✓ Installing StorageClass
 • Joining worker nodes   ...
 ✓ Joining worker nodes
Set kubectl context to "kind-cte412-02230129"
You can now use your cluster with:

kubectl cluster-info --context kind-cte412-02230129

Have a nice day!
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab
14>
```

2. Create the namespace nb-<sid> with the labels app.kubernetes.io/part-of=campus-noticeboard and cte412/practical="06", and an annotation recording your full name.

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: nb-02230129
  labels:
    app.kubernetes.io/part-of: campus-notice-board # required by brief
    cte412/practical: "06" # required by brief (string)
  annotations:
    student.fullname: "Karma Wangchuk Titung" # required by brief
```

3. In that namespace, create a ResourceQuota with these hard limits: requests.cpu 1, requests.memory 1Gi, limits.cpu 2, limits.memory 2Gi, pods 15, count/services 4.

```yaml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: nb-quota
  namespace: nb-02230129
spec:
  hard:
    requests.cpu: "1" # 6 app pods (600m) + monitor (10m)
    requests.memory: 1Gi
    limits.cpu: "2"
    limits.memory: 2Gi
    pods: "15" # 6 app + monitor + surge + terminating pod
    count/services: "4" # notice-board, notice-board-ext,
```

4. Create a LimitRange supplying default requests and limits for containers. Choose the values yourself and justify each one in a comment against the quota above.

```yaml
apiVersion: v1
kind: LimitRange
metadata:
  name: nb-defaults
  namespace: nb-02230129
spec:
  limits:
    - type: Container
      default:
        cpu: 200m # <= limits.cpu ceiling; 6
        memory: 256Mi # 6 pods × 256Mi = 1.5
      defaultRequest:
        cpu: 100m # 6 pods × 100m = 600m < r
        memory: 128Mi # 6 pods × 128Mi = 768
```

5. Make nb-<sid> the default namespace of your kubectl context.

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/00-namespace.yaml
namespace/nb-02230129 created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/01-quota.yaml
resourcequota/nb-quota created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/base/02-limitrange.yaml
limitrange/nb-defaults created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config set-context --current --namespace=nb-02230129
Context "kind-cte412-02230129" modified.
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get nodes -o wide
NAME                          STATUS   ROLES           AGE     VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION
    CONTAINER-RUNTIME
cte412-02230129-control-plane   Ready    control-plane   4m51s   v1.37.0   172.24.0.6    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd6
4)   containerd://2.3.4
cte412-02230129-worker          Ready    <none>          4m34s   v1.37.0   172.24.0.4    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd6
4)   containerd://2.3.4
cte412-02230129-worker2         Ready    <none>          4m34s   v1.37.0   172.24.0.7    <none>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd6
4)   containerd://2.3.4
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl describe quota nb-quota -n nb-02230129
Name:           nb-quota
Namespace:      nb-02230129
Resource        Used  Hard
--------        ----  ----
count/services  0     4
limits.cpu      0     2
limits.memory   0     2Gi
pods            0     15
requests.cpu    0     1
requests.memory 0     1Gi
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

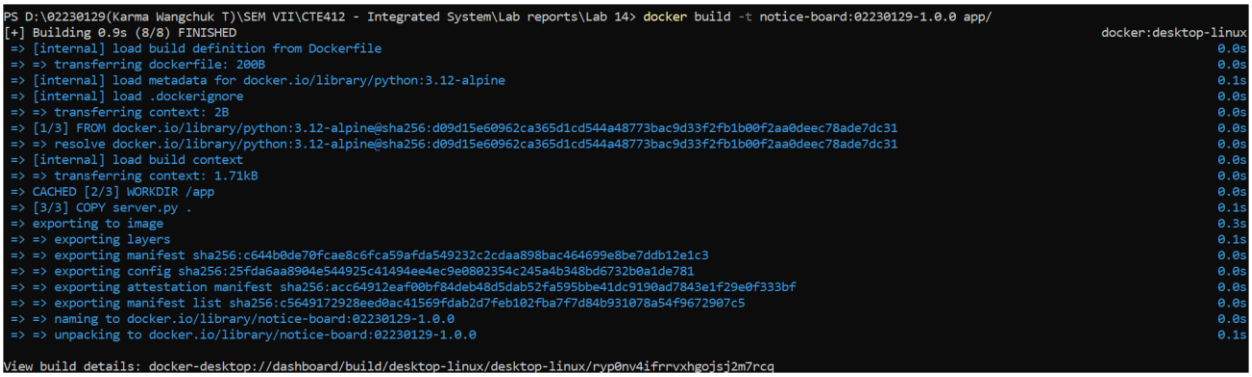
```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl config view --minify | Select-String "namespace"

    namespace: nb-02230129


PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

Task 2

1. Every version listens on port 8080, runs as a non-root user, and exposes three endpoints. GET /notices returns at least three notices as JSON. GET /version returns JSON containing version, student (your <sid>) and pod (the container hostname). GET /healthz returns HTTP 200 when healthy.
2. docker tag noticeboard-app:v1.0.0 notice-board:02230129-1.0.0
3. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
4. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
5. Tag notice-board:<sid>-1.2.0-broken — identical to 1.1.0 except that /healthz never returns 200. /notices and /version still work.
6. Place all three images into the cluster without using a registry.





```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker build -f app/Dockerfile.broken -t notice-board:02230129-1.2.0 app/
[+] Building 0.6s (8/8) FINISHED                                                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.broken                                                                                                        0.0s
 => => transferring dockerfile: 190B                                                                                                                               0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                                                                              0.0s
 => [internal] load .dockerignore                                                                                                                                  0.0s
 => => transferring context: 2B                                                                                                                                    0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:d09d15e60962ca365d1cd544a48773bac9d33f2fb1b00f2aa0deec78ade7dc31                                         0.0s
 => => resolve docker.io/library/python:3.12-alpine@sha256:d09d15e60962ca365d1cd544a48773bac9d33f2fb1b00f2aa0deec78ade7dc31                                         0.0s
 => [internal] load build context                                                                                                                                  0.0s
 => => transferring context: 1.70kB                                                                                                                                0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                                                      0.0s
 => [3/3] COPY server_broken.py server.py                                                                                                                          0.0s
 => exporting to image                                                                                                                                             0.2s
 => => exporting layers                                                                                                                                            0.1s
 => => exporting manifest sha256:120a303e4b8e59d0f17ddc007be14e3a34a876a52ce93b7512477ecd9d2c7ee5                                                                  0.0s
 => => exporting config sha256:6213d8168591b56fae65c247e53f78a05c9e4f2f538d813fa5b6261f3733ed71                                                                    0.0s
 => => exporting attestation manifest sha256:59eb681c75b15ed94bc34b034b2112e86c5135209084ed881deefdfd5d03d112                                                      0.0s
 => => exporting manifest list sha256:d0dab01c9bd7c6a9ea4bbf72096fe1656c4bbe55cb00b41a6012f0889ae3dbdb                                                             0.0s
 => => naming to docker.io/library/notice-board:02230129-1.2.0                                                                                                     0.0s
 => => unpacking to docker.io/library/notice-board:02230129-1.2.0                                                                                                  0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/p1acx0x8o25etfpipcp8okz2y
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
Administrator: Windows PowerShell
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind load docker-image notice-board:02230129-1.0.0 --name cte412-02230129
Image: "notice-board:02230129-1.0.0" with ID "sha256:c5649172928eed0ac41569fdab2d7feb102fba7f7d84b931078a54f9672907c5" not yet present on node "cte412-02230129-worker2", lo
ading...
Image: "notice-board:02230129-1.0.0" with ID "sha256:c5649172928eed0ac41569fdab2d7feb102fba7f7d84b931078a54f9672907c5" not yet present on node "cte412-02230129-control-plan
e", loading...
Image: "notice-board:02230129-1.0.0" with ID "sha256:c5649172928eed0ac41569fdab2d7feb102fba7f7d84b931078a54f9672907c5" not yet present on node "cte412-02230129-worker", loa
ding...
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind load docker-image notice-board:02230129-1.1.0 --name cte412-02230129
Image: "notice-board:02230129-1.1.0" with ID "sha256:ef54750ca456abba7911e499a42cbcb0c5ce1517555f8e86635f119417316bcf" not yet present on node "cte412-02230129-worker2", lo
ading...
Image: "notice-board:02230129-1.1.0" with ID "sha256:ef54750ca456abba7911e499a42cbcb0c5ce1517555f8e86635f119417316bcf" not yet present on node "cte412-02230129-control-plan
e", loading...
Image: "notice-board:02230129-1.1.0" with ID "sha256:ef54750ca456abba7911e499a42cbcb0c5ce1517555f8e86635f119417316bcf" not yet present on node "cte412-02230129-worker", loa
ding...
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kind load docker-image notice-board:02230129-1.2.0 --name cte412-02230129
Image: "notice-board:02230129-1.2.0" with ID "sha256:d0dab01c9bd7c6a9ea4bbf72096fe1656c4bbe55cb00b41a6012f0889ae3dbdb" not yet present on node "cte412-02230129-worker2", lo
ading...
Image: "notice-board:02230129-1.2.0" with ID "sha256:d0dab01c9bd7c6a9ea4bbf72096fe1656c4bbe55cb00b41a6012f0889ae3dbdb" not yet present on node "cte412-02230129-control-plan
e", loading...
Image: "notice-board:02230129-1.2.0" with ID "sha256:d0dab01c9bd7c6a9ea4bbf72096fe1656c4bbe55cb00b41a6012f0889ae3dbdb" not yet present on node "cte412-02230129-worker", loa
ding...
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker images --filter "reference=notice-board"
REPOSITORY     TAG               IMAGE ID       CREATED         SIZE
notice-board   02230129-1.2.0    d0dab01c9bd7   2 minutes ago   74MB
notice-board   02230129-1.0.0    c5649172928e   5 minutes ago   74MB
notice-board   02230129-1.1.0    ef54750ca456   5 minutes ago   74MB
notice-board   latest            8e918bfff2db   6 weeks ago     97.7MB
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker exec cte412-02230129-worker crictl images | Select-String notice-board

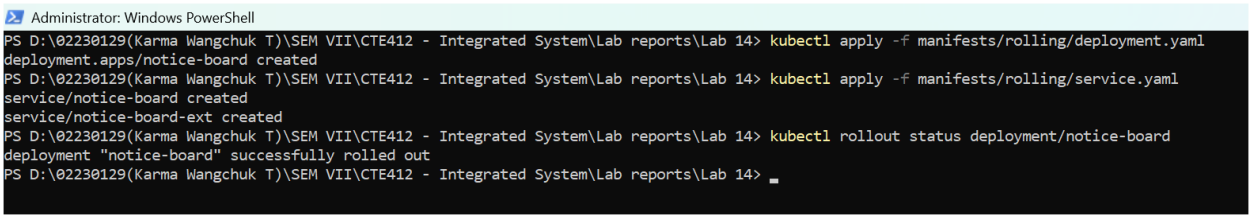
docker.io/library/notice-board                02230129-1.0.0      25fda6aa8904e       18MB
docker.io/library/notice-board                02230129-1.1.0      6d38aeafe412a       18MB
docker.io/library/notice-board                02230129-1.2.0      6213d8168591b       18MB

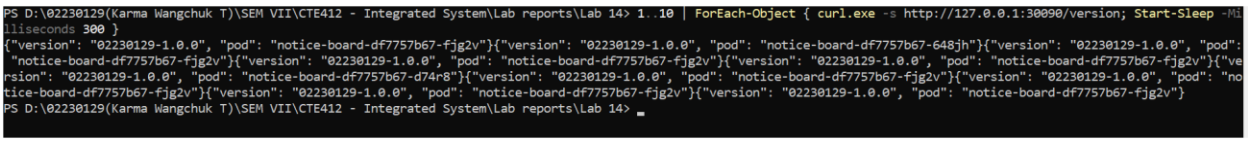

PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

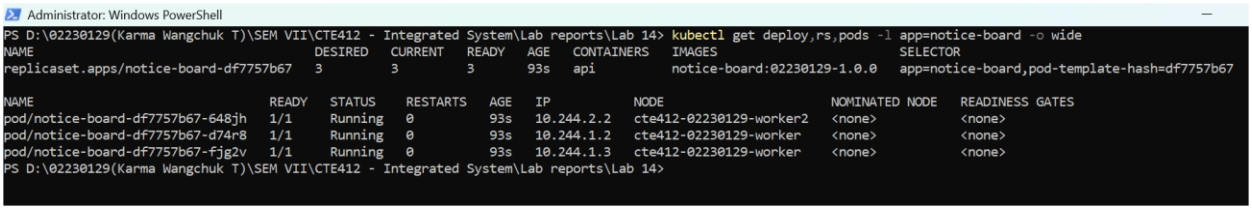
```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker run -d --name nb-test -p 8081:8080 notice-board:02230129-1.0.0
e351021736a6d1732e1d32d55093146fb77d141aeb3c5830e9bf6554e2c0f9db
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 2
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> curl.exe -s -o NUL -w "under5s: %{http_code}`n" http://127.0.0.1:8081/healthz
under5s: 503
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 4
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> curl.exe -s -o NUL -w "over5s: %{http_code}`n"  http://127.0.0.1:8081/healthz
over5s: 200
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> curl.exe -s http://127.0.0.1:8081/version
{"version": "02230129-1.0.0", "pod": "e351021736a6"}
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> docker rm -f nb-test
nb-test
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

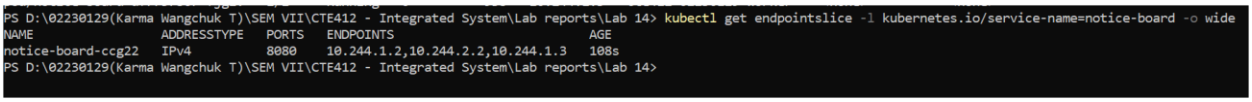
Task 3 — Deploy and Expose Version 1.0.0 (25 minutes)

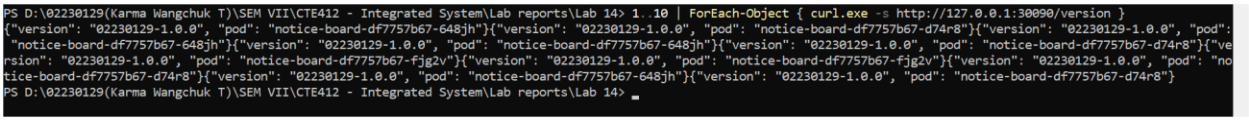
1. Create a Deployment notice-board running 3 replicas of 1.0.0, with labels app=notice-board and version=<image version> on the Pods.
2. Declare requests and limits for the container that fit inside your quota with room for Tasks 5 to 8.
3. Configure a readiness probe against /healthz. Configure a liveness probe that does not use /healthz; state in a comment why.
4. Ensure the kubelet can never attempt a registry pull for this image.
5. Create a ClusterIP Service notice-board (port 80 to the named container port) and a NodePort. Service notice-board-ext on node port 30090.
6. From the host, show that /version on port 30090 is answered by at least two different Pods.







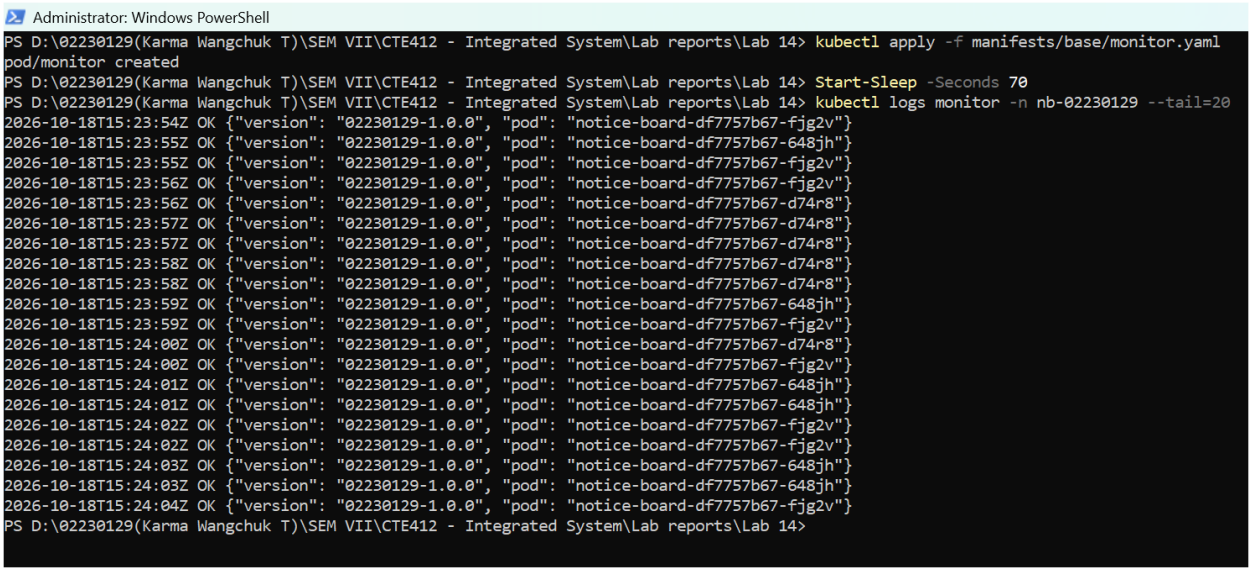


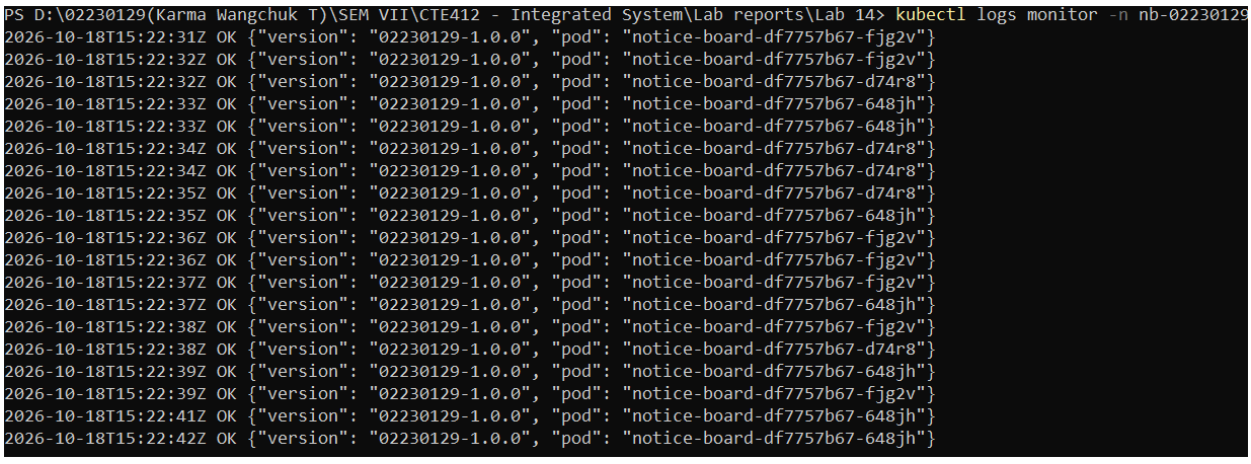


Task 4 — Availability Monitor and Summariser (15 minutes)

The monitor is the measuring instrument for Tasks 5 to 8. It must run continuously and must not be restarted between measurements.

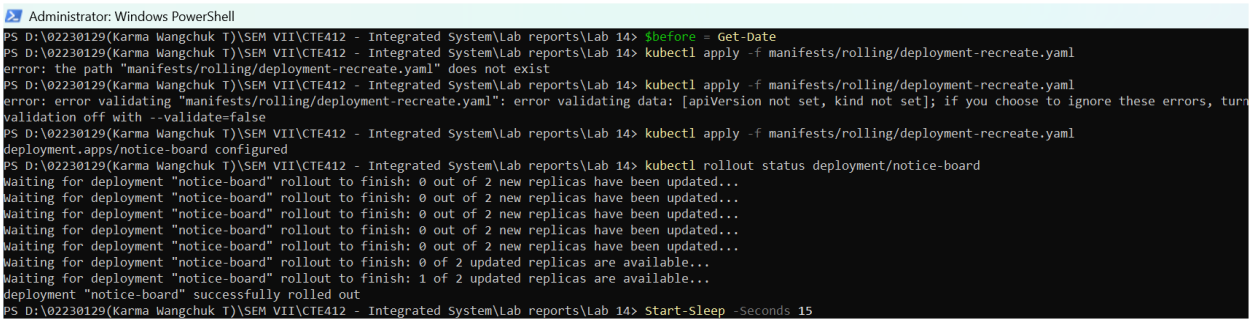
1. Create a Pod monitor in manifests/base/ that requests http://notice-board/version every 0.5 seconds, forever, and writes one line per request to standard output in the form <ISO-8601 timestamp> <OK|FAIL> <response body or error>.
2. Write scripts/summarise.sh. Given a saved monitor log as its only argument, it prints: total requests, failed requests, the first and last timestamp, and the count of responses per distinct version value.
3. Save one minute of monitor output with the system at rest and summarize it.

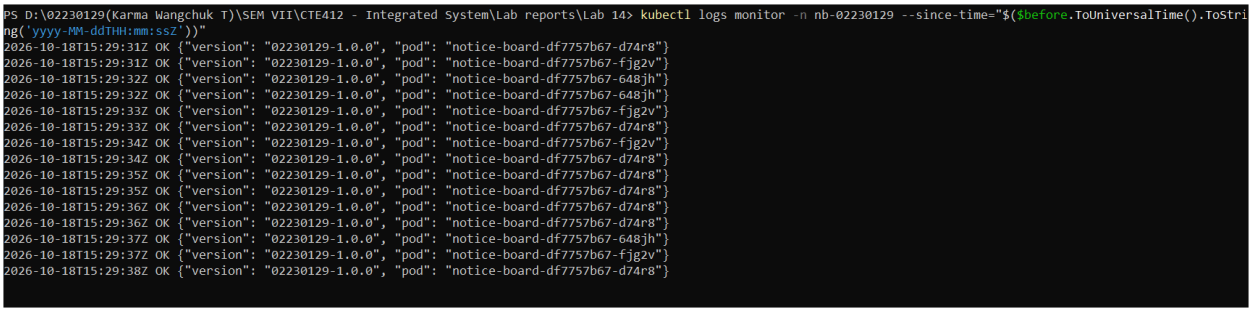


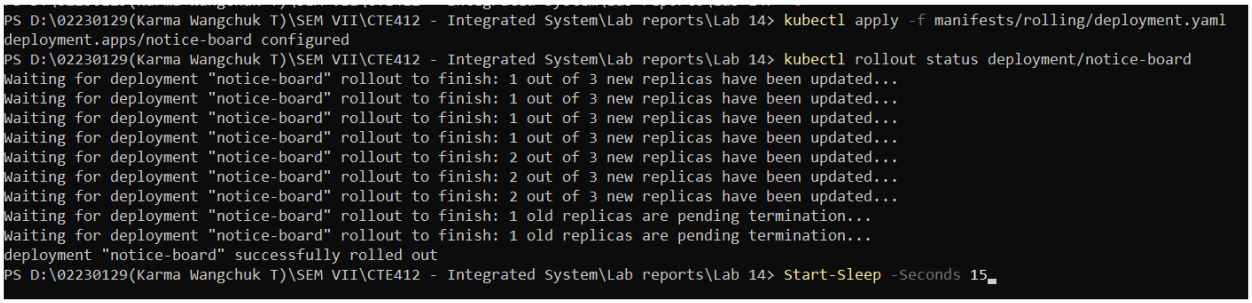


Task 5

1. Set the Deployment strategy to Recreate. Release 1.1.0 by editing and applying the manifest. Record a change-cause for the revision.
2. Return to 1.0.0 declaratively.
3. Set the strategy to RollingUpdate with at most zero unavailable Pods, at most one surge Pod, and a minimum ready time of 5 seconds. Release 1.1.0 again. Record a change-cause.
4. From each log, identify the mixed-version window: the interval during which both 1.0.0 and 1.1.0 responses were observed.







```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> $before2 = Get-Date
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Apply the updated manifest
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/rolling/deployment-rolling.yaml
deployment.apps/notice-board configured
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Monitor the rollout progress
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
deployment "notice-board" successfully rolled out
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> # Wait 15 seconds to collect logs
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 15
```

Task 6

1. Set the Deployment so that a rollout making no progress for 60 seconds is marked failed. Release 1.2.0-broken declaratively, with a change-cause.
2. Show that Kubernetes itself reported the rollout as failed, and state the condition type and reason it reported.
3. Roll back to the previous revision. This is the only imperative state change permitted in the practical.
4. Bring the repository manifest back into agreement with the live Deployment, and prove that they agree.
5. The monitor must record zero failures between the broken apply and the completed rollback. If it records any, your configuration is wrong: correct it and repeat the task.

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> $t0 = (Get-Date).ToUniversalTime().ToString("yyyy-MM-ddTHH:mm:ssZ")
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/rolling/deployment-broken.yaml
deployment.apps/notice-board configured
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Start-Sleep -Seconds 70
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get deployment notice-board -o jsonpath="{range .status.conditions[*]}{.type
}={.status} reason={.reason}{'\n'}{end}"
Available=True reason=MinimumReplicasAvailable
Progressing=False reason=ProgressDeadlineExceeded
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get pods -l app=notice-board -o wide
NAME                            READY   STATUS    RESTARTS   AGE     IP           NODE                      NOMINATED NODE   READINESS GATES
notice-board-8fb7f7bb5-w94qf    0/1     Running   0          2m4s    10.244.2.8   cte412-02230129-worker2   <none>           <none>
notice-board-fcf5-f9z6f         1/1     Running   0          5m34s   10.244.1.7   cte412-02230129-worker    <none>           <none>
notice-board-fcf5-k7gtf         1/1     Running   0          5m28s   10.244.2.7   cte412-02230129-worker2   <none>           <none>
notice-board-fcf5-srmqc         1/1     Running   0          5m23s   10.244.1.8   cte412-02230129-worker    <none>           <none>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout undo deployment/notice-board       # the ONE permitted imperative s
tep
deployment.apps/notice-board rolled back
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board
deployment "notice-board" successfully rolled out
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
deployment "notice-board" successfully rolled out
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl logs monitor -n nb-02230129 --since-time=$t0
2026-10-18T15:40:55Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:40:56Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:40:58Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:40:59Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:03Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:04Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:06Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:07Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
2026-10-18T15:41:09Z FAIL wget: can't connect to remote host (10.96.193.190): Connection refused
```

Task 7

1. In manifests/bluegreen/, create Deployments notice-board-blue (1.0.0) and notice-boardgreen (1.1.0), each with 3 replicas, the same probes and resources as Task 3, and a Pod label slot with value blue or green.
2. The production Services (notice-board and notice-board-ext) send traffic to blue only. A third Service, notice-board-preview, always sends traffic to green.
3. Verify green through the preview Service from inside the cluster while production still serves blue.
4. Switch production to green by editing and applying the Service manifest(s). Then switch back to blue, then to green again.
5. Capture the monitor log across all three switches.

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl delete -f manifests/rolling/deployment.yaml
deployment.apps "notice-board" deleted
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
Administrator: Windows PowerShell
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl delete -f manifests/rolling/deployment.yaml
deployment.apps "notice-board" deleted
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/bluegreen/blue.yaml
deployment.apps/notice-board-blue created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/bluegreen/green.yaml
deployment.apps/notice-board-green created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/bluegreen/flags.yaml
error: the path "manifests/bluegreen/flags.yaml" does not exist
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl apply -f manifests/bluegreen/services.yaml
service/notice-board configured
service/notice-board-ext configured
service/notice-board-preview created
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board-blue
Waiting for deployment "notice-board-blue" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "notice-board-blue" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "notice-board-blue" rollout to finish: 2 of 3 updated replicas are available...
deployment "notice-board-blue" successfully rolled out
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl rollout status deployment/notice-board-green
Waiting for deployment "notice-board-green" rollout to finish: 0 of 3 updated replicas are available...
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl run curl-preview --image=busybox:1.36 --rm -it --restart=Never -- wget -qO-
http://notice-board-preview/version
Error from server (Forbidden): pods "curl-preview" is forbidden: exceeded quota: nb-quota, requested: limits.cpu=200m, used: limits.cpu=2, limited: limits.cpu=2
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> curl.exe -s http://127.0.0.1:30090/version
{"version": "02230129-1.0.0", "pod": "notice-board-blue-b87d475d9-xw9x9"}
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> Get-Date -Format "yyyy-MM-ddTHH:mm:ssZ"
2026-10-18T21:53:47Z
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get endpointslice -l kubernetes.io/service-name=notice-board -o wide
NAME                 ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
notice-board-ndkls   IPv4          8080    10.244.1.9,10.244.1.10,10.244.2.9   16m
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get endpointslice -l kubernetes.io/service-name=notice-board-preview -o wide

NAME                        ADDRESSTYPE   PORTS   ENDPOINTS                                 AGE
notice-board-preview-gzqb7   IPv4          8080    10.244.1.11,10.244.2.11,10.244.2.10   4m51s
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get endpointslice -l kubernetes.io/service-name=notice-board -o wide
NAME                ADDRESSTYPE   PORTS   ENDPOINTS                         AGE
notice-board-ndkls   IPv4          8080    10.244.1.9,10.244.1.10,10.244.2.9   17m
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl get endpointslice -l kubernetes.io/service-name=notice-board-preview -o wide

NAME                        ADDRESSTYPE   PORTS   ENDPOINTS                                 AGE
notice-board-preview-gzqb7   IPv4          8080    10.244.1.11,10.244.2.11,10.244.2.10   5m44s
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

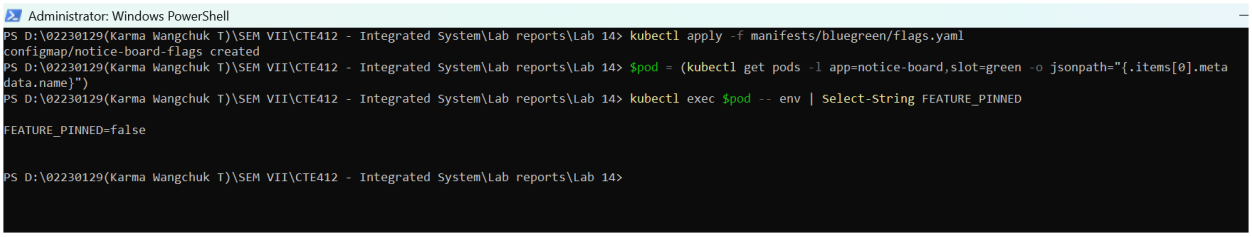
```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl logs monitor -n nb-02230129
2026-10-18T15:22:31Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:32Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:32Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:33Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:33Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:34Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:34Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:35Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:35Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:36Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:36Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:37Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:37Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:38Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
2026-10-18T15:22:38Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-d74r8"}
2026-10-18T15:22:39Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-648jh"}
2026-10-18T15:22:39Z OK {"version": "02230129-1.0.0", "pod": "notice-board-df7757b67-fjg2v"}
```

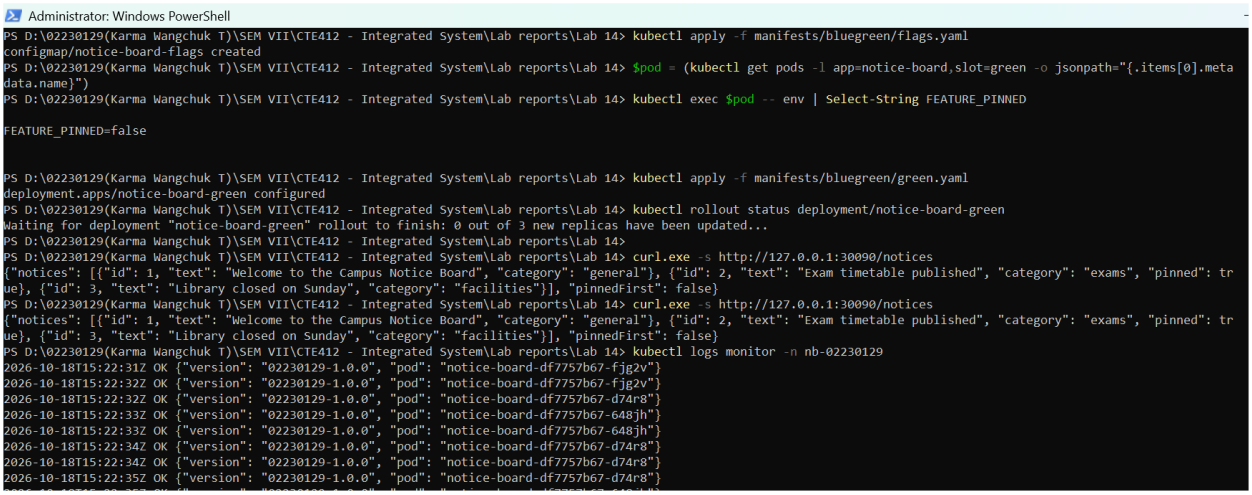
Administrator: Windows PowerShell

```
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14> kubectl describe quota nb-quota -n nb-02230129
Name:            nb-quota
Namespace:       nb-02230129
Resource         Used   Hard
--------         ----   ----
count/services   3      4
limits.cpu       2      2
limits.memory    1Gi    2Gi
pods             7      15
requests.cpu     610m   1
requests.memory  400Mi  1Gi
PS D:\02230129(Karma Wangchuk T)\SEM VII\CTE412 - Integrated System\Lab reports\Lab 14>
```

Task 8

1. Create a ConfigMap notice-board-flags holding FEATURE_PINNED with the value false, and make the green Deployment read the variable from it.
2. Change the flag to true in the manifest and apply it. Determine, with evidence, whether the running green Pods observe the change.
3. Make the new flag value take effect on all green Pods with zero failed requests, using only declarative changes.





Task 9

1. Save the complete monitor log for the session as evidence/t9-monitor-full.log.
2. Complete the README rebuild sequence, then prove it: delete every object in nb-<sid> except the namespace, quota and limit range, and recreate the final state of Task 8 using the README commands only.
3. Reset the context namespace to default, delete the cluster, and commit.